

0.1 Singular Value Decomposition (SVD) and Geometric Projections

While the Spectral Theorem strictly applies to self-adjoint operators on a single vector space, the Singular Value Decomposition (SVD) generalizes this geometric decoupling to arbitrary linear maps between vector spaces of potentially different dimensions.

Let V and W be finite-dimensional inner product spaces over F (where F is R or C). Consider a linear map $T : V \rightarrow W$. Its adjoint $T^* : W \rightarrow V$ is uniquely defined by the relation $\langle T(v), w \rangle_W = \langle v, T^*(w) \rangle_V$ for all $v \in V$ and $w \in W$.

The composite operator $T^*T : V \rightarrow V$ is self-adjoint and positive semi-definite. Consequently, its eigenvalues are real and non-negative.

To prove self-adjointness, we take the adjoint of the composition. By the properties of adjoints, $(T^*T)^* = T^*(T^*)^* = T^*T$. Thus, T^*T is self-adjoint.

To prove it is positive semi-definite, let $v \in V$. We evaluate the inner product:

$$\langle T^*T(v), v \rangle_V = \langle T(v), T(v) \rangle_W = \|T(v)\|_W^2$$

By the axioms of norms, $\|T(v)\|_W^2 \geq 0$. If λ is an eigenvalue of T^*T with corresponding unit eigenvector v , then $T^*T(v) = \lambda v$, which implies:

$$\lambda = \lambda \langle v, v \rangle_V = \langle \lambda v, v \rangle_V = \langle T^*T(v), v \rangle_V = \|T(v)\|_W^2 \geq 0$$

Thus, all eigenvalues of T^*T are real and non-negative.

Let $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0$ be the eigenvalues of T^*T , counted with multiplicity. The **singular values** of T , denoted σ_i , are the non-negative square roots of these eigenvalues: $\sigma_i = \sqrt{\lambda_i}$.

[Singular Value Decomposition] Let $T : V \rightarrow W$ be a linear map with rank r . There exist orthonormal bases $\{v_1, \dots, v_n\}$ of V and $\{u_1, \dots, u_m\}$ of W , and strictly positive singular values $\sigma_1 \geq \dots \geq \sigma_r > 0$, such that for all $v \in V$:

$$T(v) = \sum_{i=1}^r \sigma_i \langle v, v_i \rangle_V u_i$$

Because T^*T is a self-adjoint operator on the finite-dimensional space V , the Spectral Theorem (Theorem 3.2.3) guarantees the existence of an orthonormal basis $\{v_1, \dots, v_n\}$ of V consisting of eigenvectors of T^*T . Order these eigenvectors such that their corresponding eigenvalues satisfy $\lambda_1 \geq \dots \geq \lambda_r > 0$ and $\lambda_{r+1} = \dots = \lambda_n = 0$. The singular values are $\sigma_i = \sqrt{\lambda_i}$.

For $1 \leq i \leq r$, define vectors $u_i \in W$ by:

$$u_i = \frac{1}{\sigma_i} T(v_i)$$

We first demonstrate that the set $\{u_1, \dots, u_r\}$ is orthonormal in W . For any $1 \leq i, j \leq r$, we compute the inner product:

$$\langle u_i, u_j \rangle_W = \left\langle \frac{1}{\sigma_i} T(v_i), \frac{1}{\sigma_j} T(v_j) \right\rangle_W = \frac{1}{\sigma_i \sigma_j} \langle T^*T(v_i), v_j \rangle_V$$

Since v_i is an eigenvector of T^*T with eigenvalue $\lambda_i = \sigma_i^2$, this becomes:

$$\langle u_i, u_j \rangle_W = \frac{\sigma_i^2}{\sigma_i \sigma_j} \langle v_i, v_j \rangle_V$$

Because $\{v_1, \dots, v_n\}$ is an orthonormal basis, $\langle v_i, v_j \rangle_V = \delta_{ij}$ (the Kronecker delta). Thus, $\langle u_i, u_j \rangle_W = \delta_{ij}$, proving $\{u_1, \dots, u_r\}$ is an orthonormal set. We can extend this set to a full orthonormal basis $\{u_1, \dots, u_m\}$ for W using the Gram-Schmidt process.

To verify the decomposition, let $v \in V$. We express v in terms of the orthonormal basis of V :

$$v = \sum_{i=1}^n \langle v, v_i \rangle_V v_i$$

Applying the linear map T yields:

$$T(v) = \sum_{i=1}^n \langle v, v_i \rangle_V T(v_i)$$

For $i > r$, $\lambda_i = 0$, which implies $\|T(v_i)\|_W^2 = \langle T^*T(v_i), v_i \rangle_V = 0$, forcing $T(v_i) = 0_W$. Therefore, the sum truncates at r . Substituting $T(v_i) = \sigma_i u_i$ for $1 \leq i \leq r$ gives the desired representation:

$$T(v) = \sum_{i=1}^r \sigma_i \langle v, v_i \rangle_V u_i$$